

Lab: Synaptic Plasticity and Learning Circuits

Learning Goal: Trace learning mechanisms from molecular synaptic changes to behavioral outcomes.

Part 1: Hebbian Learning Scenarios

Exercise: For each scenario, identify what neural connections are being strengthened or weakened, and whether this is LTP or LTD:

1. A child burns their hand on a hot stove and learns to avoid it
2. You stop flinching at a door that used to slam but was fixed
3. A musician practices a difficult passage until it becomes automatic
4. A phobia develops after a traumatic experience

Write your response here

Part 2: Dopamine Prediction Error Analysis

Learning Goal: Trace dopamine signals through reward learning scenarios.

For each scenario, predict the dopamine response (burst, baseline, or dip):

Scenario	Dopamine Response	Why?
You find \$20 on the ground unexpectedly		
You receive your expected paycheck		
The vending machine takes your money but gives no snack		
You win a bet you thought you'd lose		
Your favorite show is cancelled unexpectedly		

Write your response here

Part 3: Hippocampal Memory Consolidation

Learning Goal: Understand the hippocampal-cortical consolidation process.

Scenario: You visit a new city for the first time.

1. During the day: What does the hippocampus encode? (events, spatial layout, emotional associations)
2. That night during sleep: What is replayed during sharp-wave ripples?
3. Over subsequent weeks: How does the memory transfer from hippocampus to cortex?
4. After a year: Can you recall the same details? What has been lost vs. retained? Why?

Write your response here

Part 4: STDP and Prediction

Learning Goal: Connect spike-timing plasticity to prediction at the synaptic level.

Thought experiment: Neuron A fires 10ms before Neuron B, repeatedly.

1. What happens to the $A \rightarrow B$ synapse? (LTP or LTD?)
2. Now reverse: B fires 10ms before A. What happens to the $A \rightarrow B$ synapse?
3. Why does this rule make the synapse a predictor? What is A "predicting"?
4. How might this simple rule, operating across billions of synapses, implement a generative model?

Write your response here

Part 5: Reflection

In 150 words, reflect: Addiction can be understood as dopamine-driven learning gone wrong — drugs hijack the reward prediction error signal. If learning is inference, does this mean addiction is a form of "false inference"? What makes it so hard to reverse?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Scenario analysis	Hebbian LTP/LTD
2	Signal prediction	Dopamine prediction errors
3	Consolidation tracing	Hippocampal replay and sleep
4	Mechanistic reasoning	STDP as predictive learning
5	Clinical reflection	Addiction as aberrant learning